

Muhammad Rehan

Software Engineer

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Work Experience

Divine Bits (Pvt) Ltd

Islamabad, Pakistan

ML & AI Engineer Intern

June 2024 – Oct 2024

- Built an end-to-end AI-powered recipe recommendation platform suggesting recipes based on available ingredients or target cuisine.
- Implemented an ingredient-based recommender using TF-IDF and Nearest Neighbors to find recipes with high ingredient overlap.
- Developed a cuisine-based recommender using TF-IDF and Cosine Similarity on combined recipe text.
- Integrated a YOLOv8 computer vision model to detect ingredients from uploaded images for automatic recipe suggestions.
- Deployed an interactive web app using Streamlit with ingredient input, image upload, and mode toggle (Ingredient/Cuisine).

OntriggerJoy Inc.

Islamabad, Pakistan

Game Developer Intern

June 2024 – Oct 2024

- Contributed to the development of 2D and 3D games using Unity and C#, focusing on gameplay mechanics, UI, and performance optimization.
- Collaborated with a small agile team leveraging Git for version control and smooth workflow management.
- Gained hands-on experience in debugging, testing, and improving user experience for live game projects.
- Strengthened problem-solving, teamwork, and project management skills in a fast-paced development environment.

Academic Projects

Hyper-Local Disaster & Safety Network (HLDSN) | Final Year Project | Java, Firebase, P2P Mesh

- Developed a mobile application enabling resilient P2P communication and emergency coordination when conventional internet infrastructure fails.
- Implemented mesh networking for offline connectivity, supporting real-time personal chat, SOS alerts with one-tap triggering, and community incident reporting.
- Integrated offline hazard maps with color-coded overlays, government disaster alerts filtered by location, and a Safety & First Aid knowledge base accessible without internet.
- Built role-based authentication via Firebase Auth with separate dashboards for general users, volunteers, and NGOs, and a community feed for live incident updates.

FaceDetectionApp | Android, TensorFlow Lite, Java

- Developed an Android application for managing student attendance using TensorFlow Lite for real-time face recognition.

Ghost-Buster | Unity, C#, URP

- Built a dynamic 2D action-platformer using Unity featuring advanced combat mechanics, strategic enemy AI, and URP-powered environments.

SRE-Homes | MongoDB, Express, ReactJS, NodeJS (MERN)

- Developed a full-stack MERN real estate application with a seamless property browsing experience and a robust administrative dashboard for inventory management.

Technical Skills

Frontend Development: ReactJS, JavaScript, Bootstrap, HTML, CSS, Figma

Backend Development: Java, Python, Express, NodeJS, PHP, C#, C++

Databases: MongoDB, SQL, Firebase

Tools & Platforms: Git, Unity, Android Studio

Soft Skills: Team Player, Bias for Action, Deliver Results

Education

Capital University of Science and Technology

Islamabad, Pakistan

Bachelor of Science in Software Engineering

2022 – Present

Dr. Abdul Qadeer Khan School and College of Science and Technology

Rawalpindi, Pakistan

FSC Pre-Engineering

2020 – 2022